Project Progress Report
Data Warehouse

S1 Sains Data, Fakultas Matematika Ilmu Pengetahuan dan IPA
Universitas Negeri Surabaya

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Project General Information

Title	Design and Implementation of a Cryptocurrency Data Warehouse for Volatility and Sentiment-Driven Market Analysis
Topic	Financial
Revisi?	Yes / No

Description of problem formulation	The cryptocurrency market operates continuously around the clock, generating massive streams of price and transaction data without interruption. This creates both an opportunity and a challenge: the data is abundant, but deriving structured, queryable insight from it requires a proper data infrastructure. This project addresses the absence of a well-organised data warehouse that brings together raw market price data with macro-level market sentiment indicators. Most existing analyses treat these two data streams separately, which limits the depth of insight available. For instance, investigating whether periods of Extreme Fear (as measured by the Fear and Greed Index) tend to coincide with specific price movement patterns requires both datasets to be integrated into a single, consistent structure before any meaningful OLAP query can be performed. Beyond integration, there is also the challenge of scale and time. Hourly price data for five major crypto assets across two years produces approximately 17,520 raw rows per asset before aggregation, resulting in around 87,600 total hourly rows across all five assets. This volume demands efficient storage strategies, proper schema design, and performance-oriented database features. Without these, even basic trend analysis becomes computationally expensive. Additionally, a periodic, automated pipeline is needed to keep the warehouse up to date, allowing analysts to work with data that reflects the current state of the market rather than a static historical snapshot.
Target insight to be achieved	Through the Data Staging process (ETL pipeline) and the Data Presentation layer (OLAP), this project aims to produce five main analytical insights: 1. Volatility profile per asset and time period. By rolling up data through the time dimension and slicing by asset, the cube will reveal which of the five crypto assets (BTC, ETH, BNB, SOL, ADA) carry the highest price fluctuation risk, and during which months or quarters that risk tends to concentrate. 2. Correlation between market sentiment and price movement. Through drill-down operations on the sentiment dimension, starting from broad category labels like Fear or Greed down to precise numerical scores, the cube will expose whether extreme sentiment conditions systematically align with large positive or negative daily returns. 3. Trend distribution across the time hierarchy. By combining dice operations on the trend and asset dimensions with roll-up on the time dimension, the cube

	will show how Bullish and Bearish market regimes are distributed across coins and calendar periods. 4. Volume anomaly and its relationship to price impact. Through slice operations on the time dimension and dice operations combining the asset and sentiment dimensions, the cube will investigate whether abnormally high trading volume consistently coincides with or precedes large price swings, and under which sentiment conditions those anomalies appear most often. 5. RSI-based momentum and trend reversal patterns. Using drill-down on the time dimension to daily granularity and dice operations across the trend and asset dimensions, the cube will reveal how overbought (RSI > 70) and oversold (RSI < 30) conditions are distributed, and whether they tend to align with subsequent transitions between Bullish and Bearish trend labels
Research references	He, M., Shen, L., Zhang, Y., & Zhang, Y. (2023). Predicting cryptocurrency returns for real-world investments: A daily updated and accessible predictor. Finance Research Letters, 58, 104406. https://doi.org/10.1016/j.frl.2023.104406 Wang, J. N., Liu, H. C., & Hsu, Y. T. (2024). A U-shaped relationship between the crypto fear-greed index and the price synchronicity of cryptocurrencies. Finance Research Letters, 59, 104763. https://doi.org/10.1016/j.frl.2023.104763 Ma, F., Liang, C., Ma, Y., & Wahab, M. I. M. (2020). Cryptocurrency volatility forecasting: A Markov regime-switching MIDAS approach. Journal of Forecasting, 39(8), 1277–1290. https://doi.org/10.1002/for.2691 Sebastião, H., & Godinho, P. (2021). Forecasting and trading cryptocurrencies with machine learning under changing market conditions. Financial Innovation, 7(1), 1–30. https://doi.org/10.1186/s40854-020-00217-x Baur, D. G., Dimpfl, T., & Jung, R. C. (2024). A comparison of cryptocurrency volatility: Benchmarking new and mature asset classes. Financial Innovation, 10, 90. https://doi.org/10.1186/s40854-024-00646-y

Dataset Detail Information

Dataset source	Data acquisition was carried out through direct API pull extraction from two public sources, both accessed via HTTP requests using the Python requests library: 1. Yahoo Finance API (GET /v8/finance/chart/{symbol}): Hourly OHLCV (Open, High, Low, Close, Adjusted Close, Volume) data was retrieved for five major cryptocurrencies, namely BTC-USD, ETH-USD, BNB-USD, SOL-USD, and ADA-USD. Due to the 60-day chunk limit imposed by the Yahoo Finance API, requests are performed in 59-day chunks and then concatenated. Each asset generates an estimated 17,520 hourly rows over a two-year span, resulting in approximately 87,600 total hourly rows across all five assets. The raw results are saved locally in JSON format per chunk and then consolidated into a CSV file per symbol. 2. Alternative.me API (GET /fng/?limit=730): Daily Fear and Greed Index scores were retrieved over a two-year period, yielding approximately 730 rows. Each row contains a numerical score (0-100), its corresponding category label (Extreme Fear, Fear, Neutral, Greed, or Extreme Greed), and the event timestamp. The extraction results are saved in both JSON and CSV formats as a separate raw archive.
General description of dataset / datadump	The dataset combines hourly time-series market price data with daily granularity market sentiment data. After extraction, the hourly OHLCV data is aggregated to daily granularity during the transformation phase, producing a unified daily dataset covering a two-year historical period. The consolidated dataset includes both price-derived technical indicators (such as moving averages, RSI, and volatility) and sentiment-derived features (such as the Fear and Greed Index score and classification), forming a single source of truth for OLAP analysis.

Special aspects of the dataset	Based on an initial extraction and evaluation of historical data samples, several characteristics and specific aspects were identified: 1. Data Sparsity: The raw hourly data retrieved from Yahoo Finance has a near-perfect sparsity level of approximately 0%, since the crypto market operates 24/7 without interruption and records observations every hour. However, in the merged and feature-engineered dataset, sparsity naturally increases due to the rolling window warmup effect. For example, the 30-day Moving Average (MA30) requires a minimum of 30 prior rows before producing a valid value, and the 14-period RSI similarly requires enough historical data to initialise. At full two-year scale, this warmup-induced missingness represents a very small and diminishing proportion of the total dataset. 2. Class Imbalance: For daily trend classification, records are labelled as Bullish (MA7 > MA30), Bearish (MA7 < MA30), or Neutral (MA7 = MA30). In the sample evaluation, a moderate class imbalance was observed, with Bearish periods dominating at roughly 75.2% compared to Bullish at 24.8%, reflecting a majority-to-minority ratio of approximately 3:1. This imbalance is consistent with the fundamental nature of crypto markets, which tend to experience longer consolidation or correction phases than sharp upward rallies. 3. Outlier Characteristics and Extreme Distributions: The crypto market exhibits pronounced volatility and fat-tail risk. Based on Interquartile Range (IQR) and Z-Score detection on the sample data, roughly 20% of price (close) and volume observations fall into the anomalous category by IQR criteria. The daily_return feature showed approximately 4.4% outliers by IQR, with a small portion (around 0.5%) representing extreme deviations (|Z| > 3). Additionally, a dedicated volume_zscore indicator was computed, and observations where its absolute value exceeds 2 are flagged as is_volume_anomaly, capturing statistically abnormal trading activity.

Data Staging Progress

Extraction stage	The extraction phase has been fully implemented as the first completed phase of the ETL pipeline. Two Python modules handle each source independently, both located in src/crypto_dwh/. The first module, extract_yahoo.py, retrieves hourly OHLCV data from the Yahoo Finance v8/finance/chart/ API endpoint. Because the API enforces a 60-day limit per request, the date range is automatically divided into 59-day chunks using the _date_chunks() helper function. Each chunk is fetched, parsed into a structured DataFrame with columns for datetime_utc, symbol, open, high, low, close, adj_close, and volume, and saved locally as a JSON archive. After all chunks are fetched, duplicates are removed, results are sorted by timestamp, and the final combined DataFrame is saved as a CSV file. The function supports both an initial full historical load and an incremental mode, and implements automatic retry logic with exponential backoff for connection errors and rate-limit responses (HTTP 429). The second module, extract_fng.py, retrieves the daily Fear and Greed Index from the Alternative.me API endpoint. A single request retrieves up to 730 days of daily sentiment records. Each record includes a UNIX timestamp, a numerical score, and a category classification, which is normalised to a lowercase snake_case format (e.g., extreme_fear, greed). The parsed result is saved in both JSON and CSV formats. This module also implements the same retry and backoff logic. Both extraction modules are orchestrated by the pipeline.py module, which coordinates the extraction run, saves intermediate snapshots to data/processed/, and passes results downstream to the transformation phase. The extraction is also

	integrated into an Apache Airflow DAG (crypto_dwh_dag.py) with two separate tasks, extract_yahoo and extract_fng, scheduled to run daily at 02:00 UTC. Both tasks push XCom metadata (symbol list, row counts) to downstream tasks.
Transformation stage	The transformation phase has been fully designed and implemented in transform.py, and is currently in the integration and testing stage. The transformation consists of four sequential stages: Stage 1: Standardisation and Cleaning Hourly OHLCV data is processed by converting UNIX timestamps to UTC datetime format, casting all OHLCV columns from string to float, and removing irrelevant columns. Fear and Greed data is standardised by renaming value to fng_value and value_classification to fng_classification, and converting the score from string to integer. This stage aligns the structure and types of both sources before integration Stage 2: Integration via Aggregation and Join The cleaned hourly data is aggregated to daily granularity using the _aggregate_to_daily() function. The aggregation logic takes open from the first candle of the day, high as the daily maximum, low as the daily minimum, close from the last candle, and volume as the total sum for the day. The aggregated daily OHLCV data is then left-joined with the Fear and Greed data on the date column, producing a unified daily DataFrame per symbol. Stage 3: Feature Engineering and Indicator Computation Seven analytical indicators are computed per symbol group using the _compute_indicators() function: daily_return (percentage change in close), ma7 and ma30 (7-day and 30-day rolling means of close), volatility_7d (7-day rolling standard deviation of daily_return), rsi_14 (Wilder's exponential RSI over 14 periods using EWM), volume_zscore (Z-score of volume over a 30-day trailing window), and is_volume_anomaly (boolean flag where |volume_zscore| > 2). In addition, two categorical columns are derived: trend_label (bullish when MA7 > MA30, bearish when MA7 < MA30, neutral otherwise) and rsi_signal (oversold when RSI < 30, overbought when RSI > 70, neutral otherwise) Stage 4: Star Schema Construction The enriched daily DataFrame is decomposed into dimension tables and fact tables following the Kimball Dimensional Model. Dimension tables include dim_time (date decomposition into year, quarter, month, week, day_of_week, day_name, and is_weekend), dim_asset (symbol metadata for BTC-USD, ETH-USD, BNB-USD, SOL-USD, ADA-USD), dim_sentiment (five fixed sentiment bins mapping fng_classification to ordinal sentiment keys), and dim_trend (a dynamic combination of trend_label and rsi_signal values discovered in the data). The fact table, fact_market_daily, stores the full numerical matrix of OHLCV values, indicators, and fng_value, with foreign keys to all four dimension tables. A separate fact_market_hourly table retains the raw hourly records for each asset with foreign keys to dim_time and dim_asset. All resulting DataFrames are saved as Parquet files to data/processed/ with a run timestamp suffix, enabling efficient downstream loading without recomputation.
Load Stage	The load phase has been fully designed and partially implemented in load_supabase.py and 01_schema.sql. The target database is Supabase, a managed cloud PostgreSQL service. The DDL schema in 01_schema.sql defines all six tables: dim_time, dim_asset, dim_sentiment, dim_trend, fact_market_hourly, and fact_market_daily, each with appropriate primary keys, NOT NULL constraints, and foreign key references to

	maintain referential integrity. The load_supabase.py module connects to Supabase via its REST API using upsert operations with the Prefer: resolution=merge-duplicates header, which handles both initial loads and incremental updates without duplicating records. Tables are loaded sequentially in dependency order, with dimension tables loaded before fact tables. The upsert is performed in chunks of 500 rows per request to respect API payload limits, and includes the same retry and backoff logic applied in the extraction phase. NaN and Inf float values are cleaned to None before serialisation to ensure JSON compatibility. Three advanced PostgreSQL optimisation features have been implemented in 02_olap.sql: 1. Table Partitioning: The fact_market_hourly table is range-partitioned by date_key (an integer in YYYYMMDD format) with three annual child partitions (2024, 2025, 2026). This enables partition pruning, meaning queries filtered on a specific year or date range will only scan the relevant partition rather than the entire table. 2. Indexes: Several B-tree indexes have been created to optimise OLAP query patterns. A composite index on (asset_key, date_key) is applied to both fact tables to accelerate the most common filter pattern of slicing by asset and time range simultaneously. Additional indexes cover datetime_utc on the hourly fact table, sentiment_key and trend_key on the daily fact table, and a partial index on date_key filtered by is_volume_anomaly = true, which significantly speeds up volume anomaly queries without scanning the entire table. 3. Materialised Views: Three materialised views have been created. mv_monthly_volatility pre-computes monthly aggregates of volatility_7d, daily_return, rsi_14, and close per asset per month. mv_sentiment_vs_return pre-computes the average return, standard deviation of return, and average volatility grouped by sentiment label and asset. mv_asset_performance pre-computes overall statistics per asset including min, max, and average close price, average return, average volatility, and the number of volume anomaly days. Each view has a unique index to support efficient CONCURRENT refresh operations. The current estimated database size is approximately 50 to 60 MB for the full two-year dataset, consisting of roughly 87,600 rows in fact_market_hourly and approximately 3,650 rows in fact_market_daily

Progress of Data Presentation Work

Subject focus	The OLAP presentation layer focuses on five key analytical targets to extract insights from the dimensional schema: 1. Volatility Profiling: Slicing and rolling up daily asset volatility and returns across the calendar time hierarchy (Year, Quarter, Month) to profile risk. 2. Sentiment vs. Return Correlation: Analyzing the relationship between the daily Fear & Greed Index score/classification and asset returns to spot market patterns. 3. Trend Distribution: Evaluating the frequency of bullish, bearish, and neutral moving average trends across assets over time. 4. Volume Anomalies: Identifying trading volume anomalies (Z-Score > 2) and measuring their price and return impacts. 5. RSI Momentum & Reversal: Cross-referencing RSI signals (overbought/oversold) with trend states to identify potential trend reversal points.
Star schema	The project implements a Star Schema consisting of 2 fact tables and 4 conformed dimension tables.

Fact Tables:

- fact_market_daily — stores daily aggregated market metrics including OHLCV data, technical indicators (MA7, MA30, Volatility 7D, RSI 14, Volume Z-score), and sentiment values. This table links to all four dimensions and serves as the primary source for OLAP analysis.
- fact_market_hourly — stores high-granularity intraday price data at hourly intervals. Physically range-partitioned into 3 child tables (fact_market_hourly_2024, _2025, _2026) to optimize time-range query performance via partition pruning.

Dimension Tables:

- dim_time — time hierarchy: Year → Quarter → Month → Week → Day, enabling roll-up and drill-down operations across the time axis.
- dim_asset — asset metadata (symbol, name, category) used as the primary slice coordinate for per-asset analysis.
- dim_sentiment — Fear & Greed Index classifications (fng_label, fng_classification) used to segment and filter fact data by market sentiment.
- dim_trend — technical trend state (trend_label: bullish/bearish/neutral, rsi_signal: overbought/oversold/neutral) enabling cross-dimensional momentum analysis.

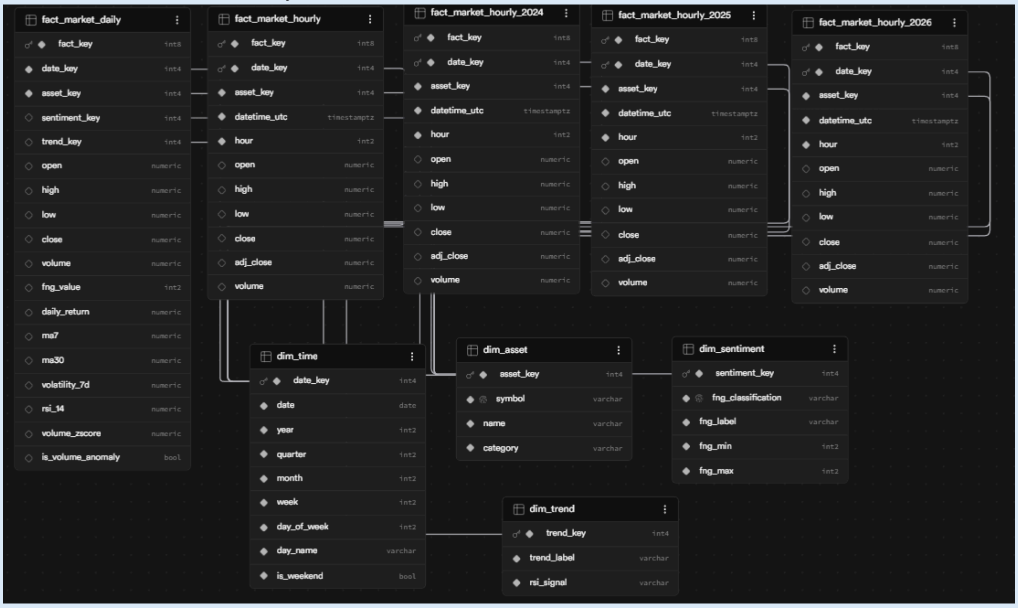

Figure 1: Physical database schema as visualized in Supabase, showing the Star Schema implementation including range-partitioned child tables of fact_market_hourly.

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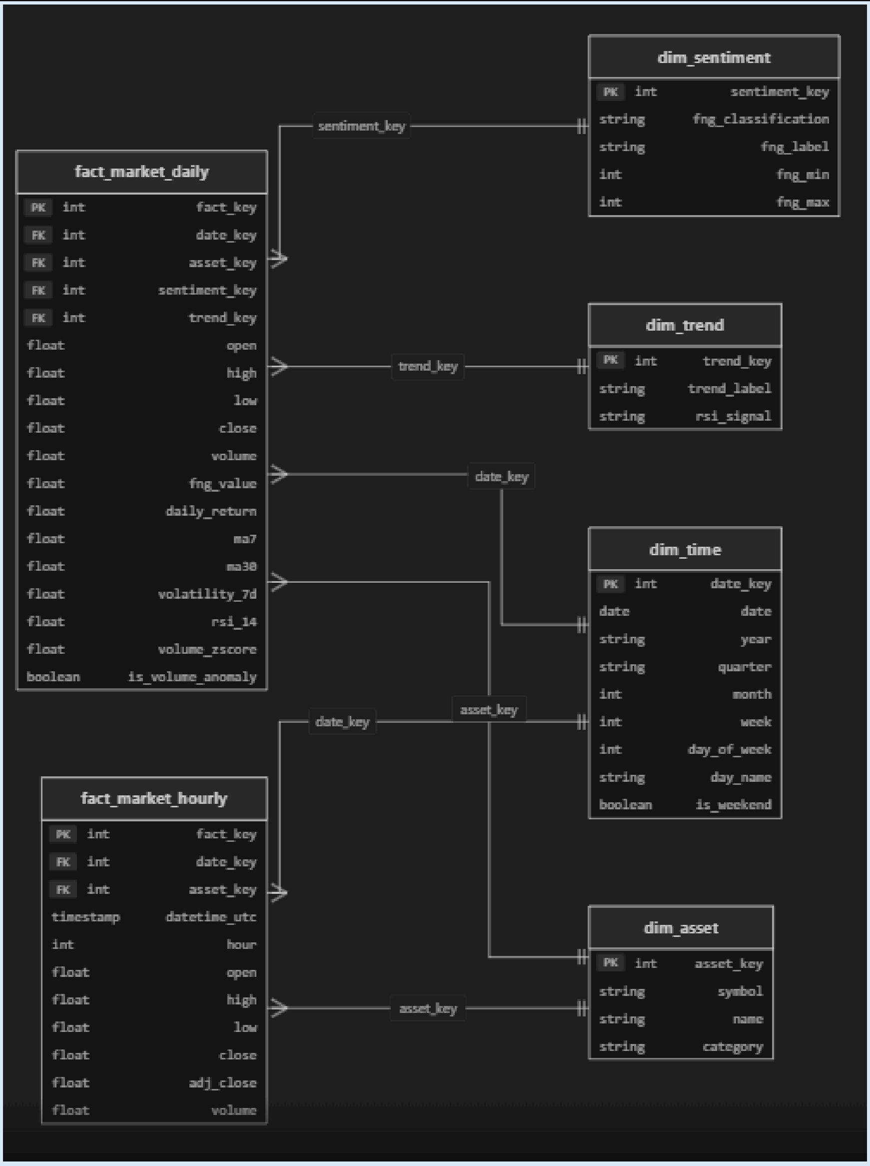

Figure 2: Logical Star Schema design showing 2 fact tables and 4 conformed dimension tables.
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<tr><td>OLAP Result</td><td>
Key multidimensional analysis results obtained from the Atoti cube and materialized views include:

1. Asset Volatility: Cardano (ADA) and Solana (SOL) exhibit the highest average 7-day volatility (rolling average of ~0.04 to 0.05), whereas Bitcoin (BTC) has the lowest (~0.02), making it the lowest-risk asset in the portfolio.
2. Sentiment Impact: Days classified as "Extreme Fear" (<25) historically align with positive daily average return rebounds, while "Extreme Greed" (>75) shows diminishing return momentum, indicating potential market tops.
3. Trend Lifespan: Slicing data by trend_label shows that BTC spent the largest percentage of days in a "bullish" moving average state compared to other assets during the 2024-2026 period.
4. Anomaly Impact: Volume anomalies (Z-score > 2) occur in less than 5% of total trading days but are strongly correlated with sharp, single-day price drops, indicating capitulation events.
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<tr><td>OLAP Visualization</td><td>
The OLAP analytical layer produces four core visualizations through the Atoti multidimensional cube, each demonstrating a distinct OLAP operation applied to the Star Schema.

1. **T1: Volatility Roll-up Table**
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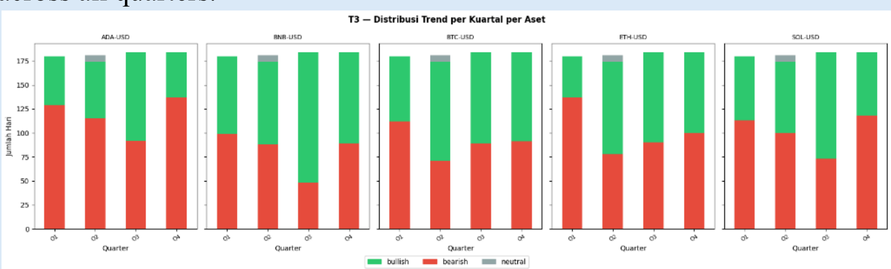
This pivot table aggregates daily volatility and returns up to the quarterly level per asset. Results show that ADA-USD and SOL-USD carry the highest risk, with average 7D volatility reaching 0.05 in Q1 and Q4, consistent with their altcoin characteristics. BTC-USD is the most stable with a flat volatility of 0.02 across Q2, Q3, and Q4. All assets show near-zero average daily returns (±0.00) at the quarterly level, indicating mean-reverting behavior across the full 2-year observation period.

symbol	quarter	Avg Volatility 7d	Avg Daily Return
ADA-USD	Q1	.05	-.00
	Q2	.03	-.00
	Q3	.03	.00
	Q4	.05	.00
BNB-USD	Q1	.03	-.00
	Q2	.02	.00
	Q3	.02	.00
	Q4	.03	.00
BTC-USD	Q1	.03	-.00
	Q2	.02	.00
	Q3	.02	.00
	Q4	.02	.00
ETH-USD	Q1	.04	-.00
	Q2	.03	.00
	Q3	.03	.00
	Q4	.03	.00
SOL-USD	Q1	.04	-.00
	Q2	.03	.00
	Q3	.04	.00
	Q4	.04	-.00

2. **T3: Trend Distribution Stacked Bar Chart**

This stacked bar chart slices data per asset and rolls up daily trend states (bullish/bearish/neutral) to quarterly counts. A dominant pattern across all 5 assets is that Q1 consistently records the highest proportion of bearish days (red), particularly visible in ADA-USD, ETH-USD, and SOL-USD. Q2 and Q3 show a recovery with bullish (green) days gaining ground. Notably, BNB-USD in Q3 shows the smallest total bar, suggesting a data-sparse period. The neutral trend (grey) remains a consistent but small minority across all quarters.

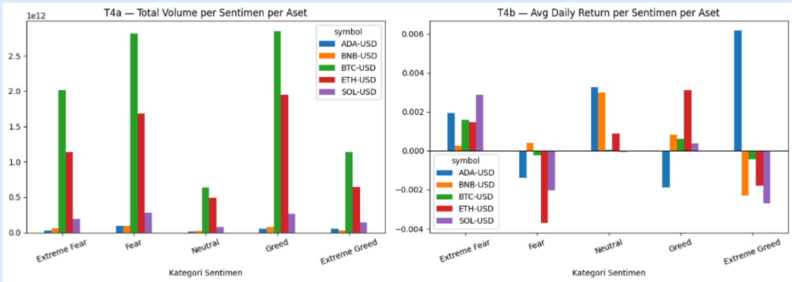


3. **T4: Volume & Return per Sentiment**

T4 presents two side-by-side charts using a Dice operation across sentiment categories and asset dimensions:

- T4a (Total Volume): BTC-USD dominates trading volume across all sentiment categories by a significant margin (reaching ~2.8 × 10^{12}

during Greed periods), confirming its position as the market's most liquid asset. Volume peaks universally occur during "Greed" sentiment periods.

- T4b (Avg Daily Return): A contrarian pattern is evident — during "Extreme Fear", most assets (ADA, BNB, SOL) record small positive average returns (rebound effect). Conversely, during "Extreme Greed", returns become negative or near-zero for BTC, ETH, and SOL, while ADA anomalously shows its highest positive return (~+0.006), suggesting different market dynamics for lower-cap assets.
- 

4. **T5: RSI Signal × Trend Label Heatmap**

   This cross-dimensional heatmap reveals three statistically significant patterns consistent across all 5 assets:

   - Oversold RSI never occurs during a bullish trend (all cells = 0). This is a definitive finding — when prices have fallen enough to trigger an oversold RSI signal, the moving average trend is always bearish or flat, never bullish.
   - Overbought RSI is almost exclusively paired with a bullish trend (ADA: 44 days, BNB: 40, BTC: 40, ETH: 49, SOL: 30 days), with near-zero occurrences in bearish states, making overbought signals a strong confirmation of bullish momentum.
   - The neutral RSI state dominates all market conditions — the highest day counts are in the neutral RSI × bearish trend (ADA: 429, ETH: 381, SOL: 363) and neutral RSI × bullish trend cells, indicating that RSI extremes are relatively rare events with high signal reliability.